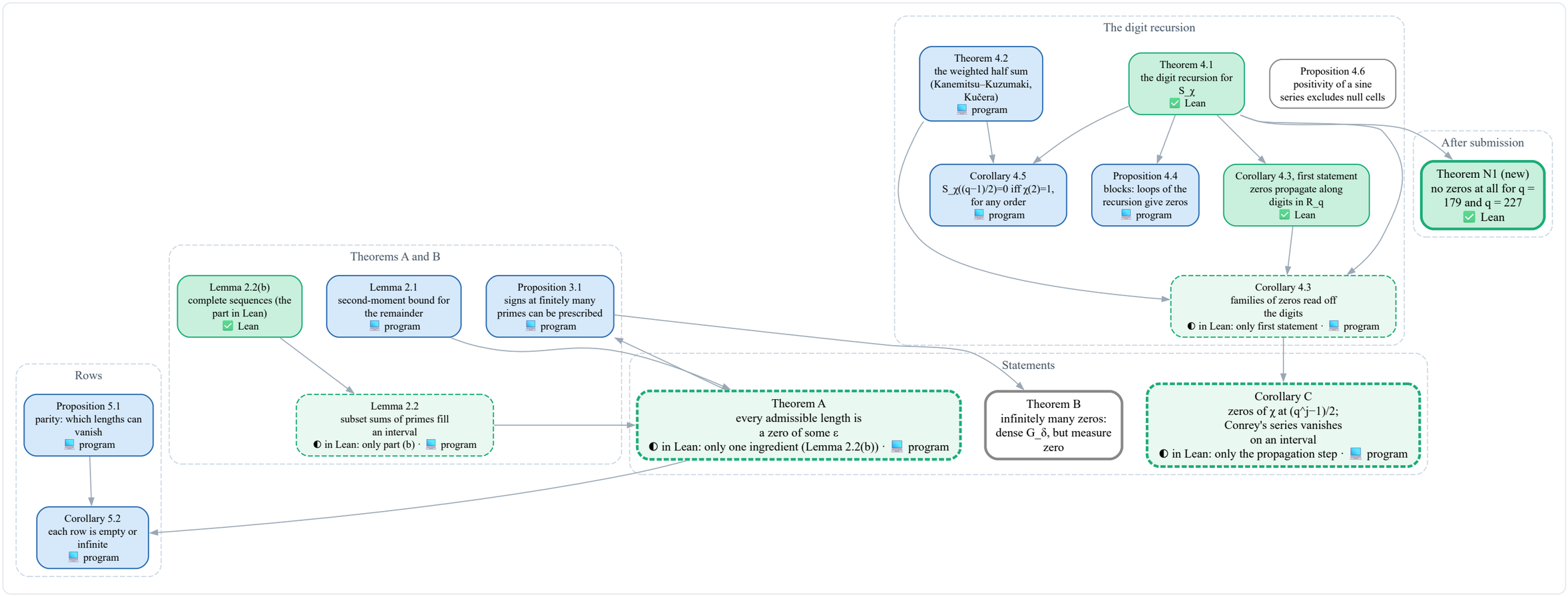
# Map of *Exact zeros of weighted sums of completely multiplicative functions*

Number theory · arithmetic functions · 11A25 · 11K65 · 11L40 · 11P70 · 11R29

**17** results · **4** formalized in Lean · **4** with a part in Lean · **11** checked by a program · written proof only · (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



## What the paper contributes

For a completely multiplicative ε with values ±1, the weighted sum $S_\varepsilon(m) = \Sigma_{n\le m}\, n\,\varepsilon(n)$ can vanish only when $m \equiv 0$ or $3 \pmod 4$.

- **Every such length is a zero** of some ε, depending on m (Theorem A), even with the signs at finitely many primes prescribed in advance, once m is large (Proposition 3.1).
- **Zeros of one fixed function are rare in measure and generic in category:** the ε with infinitely many zeros form a dense $G_\delta$ of probability zero (Theorem B). For the unweighted sums the picture is the reverse: they vanish infinitely often almost surely, by a theorem of Aymone.
- **Quadratic characters give explicit zeros.** For primes $q \equiv 7 \pmod 8$, $S_\chi((q^j-1)/2) = 0$ for every j, and Conrey's sine series of the Legendre symbol vanishes identically on an interval (Corollary C). The case j = 1 is classical (Lebesgue; Aebi–Cairns); the iteration in j and the null interval are the corollary's content. They come from a digit recursion in base q (Theorem 4.1) and the weighted half sum (Theorem 4.2, due to Kanemitsu–Kuzumaki and Kučera).
- **Added after submission:** for q = 179 and q = 227 the function $\chi_{q,1}$ has no zero at all (Theorem N1), by a finite certificate checked in Lean.

**Main tools:** a second-moment bound and Sárközy's progressions in subset sums of primes (Theorem A); the anticoncentration theorem of Sárközy and Szemerédi (Theorem B).

🚀 **Start here:** **Theorem A**: every admissible length is a zero of some ε.

**Main results:**

- **Theorem A**: every admissible length is a zero of some ε
- **Theorem B**: infinitely many zeros: dense G_δ, but measure zero
- **Corollary C**: zeros of χ at (q^j−1)/2; Conrey's series vanishes on an interval
- **Theorem N1 (new)**: no zeros at all for q = 179 and q = 227

## 🧭 Fields: where the paper lives and what it connects

**Tools taken from other branches**

- **Additive combinatorics** → arithmetic progressions in subset sums (Sárközy, with Lev's constants in the form of Bringmann–Wellnitz); complete sequences — Theorem A, Lemma 2.2
- **Probability** → a second-moment bound; the anticoncentration theorem of Sárközy and Szemerédi — Lemma 2.1, Theorem B
- **Topology (Baire category)** → dense $G_\delta$ sets in the product space of sign patterns — Theorem B
- **Algebraic number theory** → the class number formula, through the weighted half sum (Kanemitsu–Kuzumaki, Kučera) — Theorem 4.2, Corollary C
- **Formal verification** → Lean 4 + Mathlib: the digit recursion, its propagation step, complete sequences, and the finite certificate of Theorem N1 — Theorem 4.1, Corollary 4.3, first statement, Theorem N1 (new)

**What the results serve**

- **Probabilistic number theory** ← random multiplicative functions: with the weight n the picture of the unweighted sums (Aymone) is reversed — Theorem A, Theorem B
- **Analytic number theory (character sums)** ← Conrey's sine series of the Legendre symbol, from his approach to the Riemann hypothesis: it vanishes on an interval for $q \equiv 7 \pmod 8$; for $q \equiv 3 \pmod 8$ the null cells are excluded under a positivity hypothesis — Corollary C, Proposition 4.6
- **Partial sums of Legendre symbols** ← the digit recursion of Borwein–Choi–Coons and of Klurman et al., with the weight added — Theorem 4.1, Corollary 4.3